Spatial representations emerge from temporal event credit in a hippocampus-inspired navigation agent without spatial supervision

Can spatial representations emerge from sensory inputs and learning objectives that do not explicitly encode space? We investigate this question using a bottom-up computational architecture inspired by hippocampal physiology, exploring whether spatial organization can arise from the temporal structure of sensory experience and self-generated behavior, without position labels, maps, or spatial rewards.

Our model combines dentate gyrus (DG)-like sparse sensory events with CA3-like sequential memory. Temporal relationships between encountered DG events provide two complementary learning signals: a **"push"** that reinforces DG activity more strongly after longer intervals between events, and a **"pull"** that rewards shorter transitions between sensory events or, in goal-conditioned variants, faster arrival at internally defined DG targets. Together, these pressures aim to make temporal separation a more informative proxy for navigational distance. The agent learns from egocentric visual experience and intrinsic motivations, without explicit information about its physical location or spatial distances.

During exploration of a continuous visual environment, spatially informative DG activity emerges. To isolate the contribution of temporal information, we compare temporally weighted DG feedback against mean-matched constant feedback, evaluating the resulting representations on identical sensory observations. Temporal feedback produces higher DG spatial information, less redundant response maps, and a broader spatial distribution of field peaks across five paired training seeds. These improvements extend to downstream decoder representations, where spatial information also increases consistently. In contrast, temporally weighting the policy's target-hit reward does not consistently improve exploration or spatial representation quality compared with constant hit rewards.

The agent's behavior further shapes spatial coding. Even with network weights frozen, the learned policy produces more spatially informative downstream activity than random actions, demonstrating that spatial coding depends not only on the learned network but also on how the agent interacts with its environment.

Together, our results demonstrate how spatial coding can emerge from the interaction of temporal learning, sequential memory, and self-generated behavior, without explicitly optimizing for spatial information.

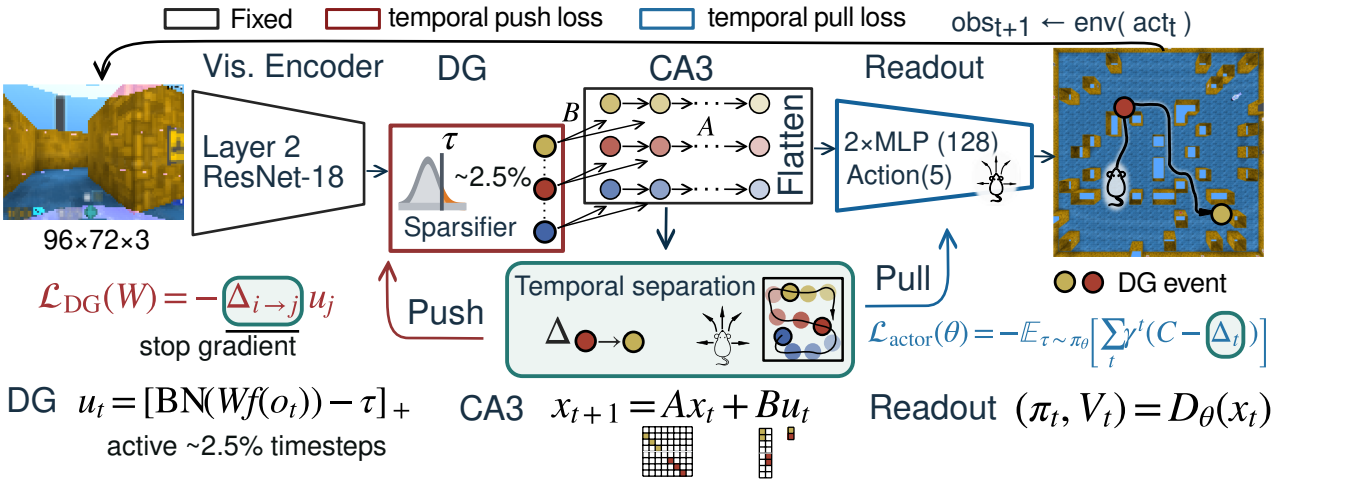


Figure 1: Push–pull architecture. BN denotes batch normalization in the DG sparsifier; the tau threshold and positive-part operation yield sparse event activity. The fixed visual trunk and CA3 memory are outlined in black; red marks the DG push objective and blue the policy pull objective. W projects visual features into DG activity, while fixed A and B shift and inject that activity into CA3 state. Delta measures the interval between eligible DG events in policy decisions, not physical distance; the timing target receives no DG gradient. The flattened CA3 state feeds a readout MLP producing a policy and value estimate. The pull equation sketches discounted policy learning; its event-timing reward differs from the goal-conditioned target-hit variants described in the text.

Architecture and learning signals. Egocentric RGB images pass through an ImageNet-pretrained ResNet-18 trunk fixed through layer 2. A trainable projection, batch normalization, and thresholded rectification then produce sparse dentate gyrus (DG) events. A fixed CA3-like shift memory retains their recent sequence. Its flattened state, together with egocentric depth, feeds a learned two-layer, 128-unit multilayer perceptron (MLP) readout that estimates value and selects among five actions. Temporal DG credit reinforces eligible event responses according to the elapsed number of decisions since a preceding event (push). Policy reward favors shorter event transitions or, in goal-conditioned variants, internally detected DG target hits (pull). C01 explores directly, C05 uses a DG target, and C15 adds frontier-based target choice. The agent receives no position coordinates, map, or spatial reward during learning (Fig. 1).

Three tests. We measure spatial coding along evolving trajectories, change actions with network weights fixed, and replay identical observations through separately trained networks (Fig. 2). These comparisons distinguish the emergence of a spatial code during learning, its expression under different behavior, and the contribution of temporal DG credit to the learned network. Common replay removes evaluation-sampling differences, although training paths may still matter.

Spatial emergence and temporal push. DG coding rises online as the agent explores (Fig. 2A). With the same 10,001 observations replayed through each trained network, temporal DG credit improves the baseline DG spatial score in all five matched seeds (mean +0.0575), reduces map overlap, and broadens the area covered by eligible field peaks (mean +0.109; Fig. 2C,D). Seed 8 illustrates the largest peak-area increase among these pairs, while panel D shows the full distribution. Readout MLP layer 1 gains spatial information in all five baseline pairs (mean +0.680 bits per activation; Fig. 2E). Its effect also has positive means for both C05 worker arms, but one of the ten C05 pairs and one of the three C15 pairs reverse direction; the frontier-goal comparison remains limited to three matched seeds.

Behavior shapes spatial coding. With each network frozen, its learned policy elicits more spatially informative readout MLP layer 1 activity and more reliable spatial maps than random actions in all 38 matched models (Fig. 2B). Thus, behavior changes the spatial signal available from a fixed network. This test identifies an effect of the agent's interaction with the environment at evaluation time; it does not by itself show how policy changes shaped the network during training.

Temporal pull. Adding temporal weighting to the C05 policy's target-hit reward has mixed effects on visited coverage and online DG spatial information when encoder credit is held fixed (Fig. 2F). Both worker bonuses reward detector hits, and actor–critic discounting already favors earlier hits. A separate longer-credit intervention jointly raises PPO discount from $\gamma = 0.99$ to 0.999 and GAE λ from 0.95 to 0.99. Coverage AUC rises in all three prescribed-goal and all three C15 seed pairs, whereas frozen-rollout learned-unit spatial information falls in all six. Those frozen evaluations use separate policy-driven paths, so they do not isolate a fixed-input change in DG coding. Detector hits and coverage do not establish physical goal arrival; matched-command and physical-arrival analyses are needed to assess goal control.

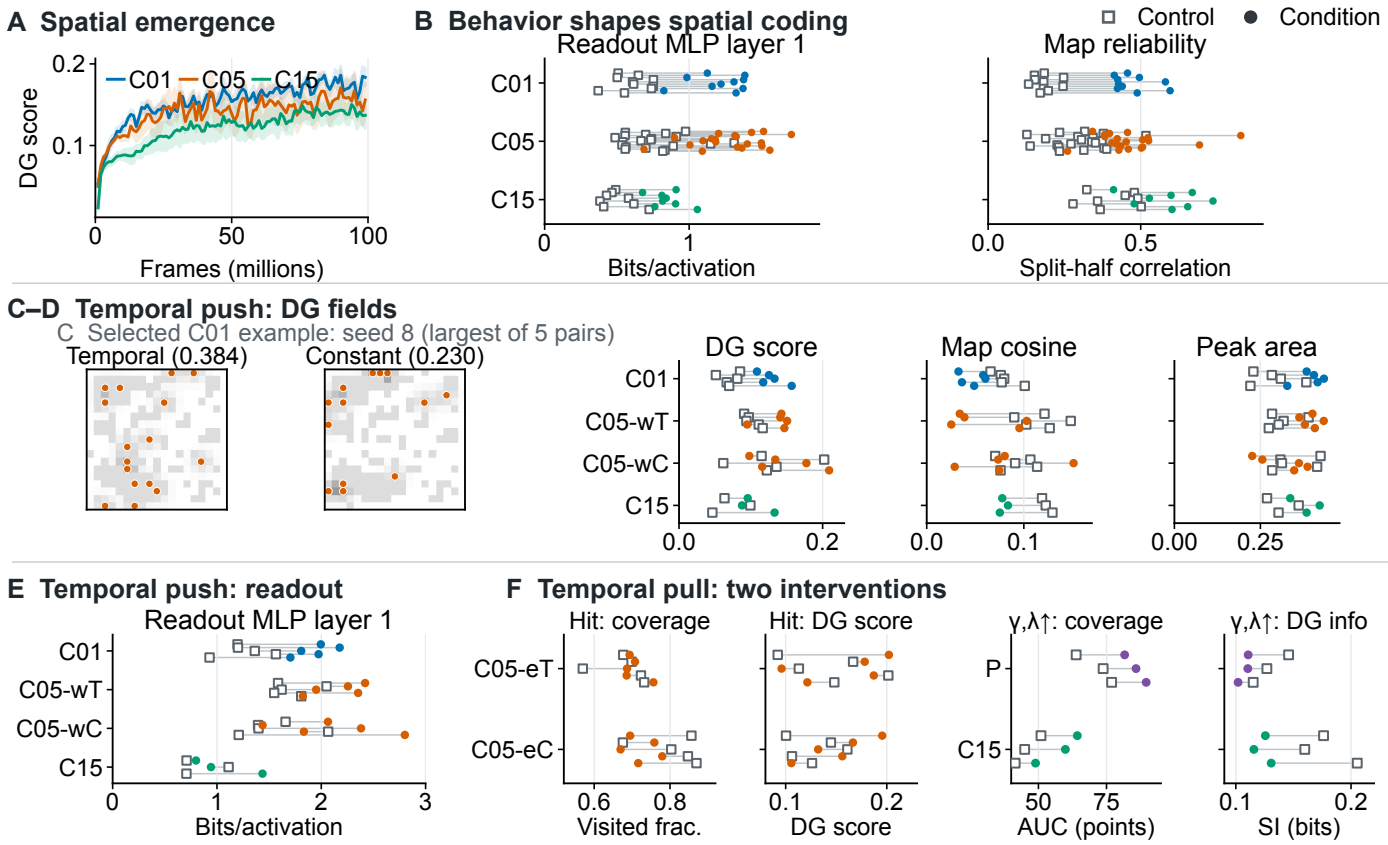


Figure 2: Temporal credit, behavior, and spatial coding. Blue, orange, green, and purple denote baseline (C01), goal-conditioned (C05), frontier-goal (C15), and prescribed-goal (P) agents. Open gray squares are controls, filled circles the tested condition, and gray lines connect matched training seeds or evaluations of the same frozen model. Each panel states its own manipulation; absolute values on differently scaled axes should not be compared across panels. **A,** Online amplitude-weighted DG spatial score over active units on each policy's latest 10,000 observations, plotted against millions of training environment frames: five-seed mean ±1 s.e.m. All curves use temporal DG credit; C05/C15 also use temporal worker hit credit. These policy-sampled measurements combine network and behavior changes over training. **B,** Frozen-model readout MLP layer 1 spatial information and split-half map correlation under random actions (gray control) versus the learned policy (colored condition). Each pair represents one of 38 terminal models, averaged over two 50,000-decision evaluations. Information is in bits per activation on equally weighted shared bins and jointly eligible units; correlation uses bins visited at least ten times in each episode-disjoint half of both policies. Both readouts favor the learned policy in all 38 models at this threshold. **C,** Baseline seed 8, selected for the largest peak-area difference among five paired seeds: temporal 0.384 versus constant 0.230 on the same 10,001 observations. Dots are DG-unit peaks after a 0.25% activity cutoff; grayscale shows Gaussian-smoothed peak density, and light gray marks unvisited bins. Effective peak area is exponential entropy of equal-mass peak density divided by 242 visited bins (100-unit Gaussian width). **D,** Temporal DG credit versus mean-matched constant credit on the identical observation sequence at terminal checkpoints. Rows hold policy and worker bonus fixed: C01 (n = 5 seed pairs), C05 with temporal or constant worker hit bonus (C05-wT/wC; n = 5 each), and C15 (n = 3). DG score is amplitude-weighted spatial information; lower mean cosine among active-unit maps indicates less overlap, and peak area is defined in C. The wT/wC suffix identifies the fixed worker bonus, not the DG-credit comparison on the horizontal axis. **E,** Readout MLP layer 1 information per activation on the same replay, with temporal versus constant DG credit and the worker arm fixed as in D. Field-eligible units have at least 20 positive observations across at least three spatial bins; each independently trained model has its own eligible-unit set. Temporal-minus-constant means are +0.680, +0.438, +0.559, and +0.217 bits per activation for C01, C05-wT, C05-wC, and C15; the numbers of positive pairs are 5/5, 5/5, 4/5, and 2/3, respectively. This normalized measure differs from an all-unit, amplitude-weighted bits-per-step score. **F,** Two distinct tests of temporal pull. The left two plots compare C05 temporal versus mean-matched constant worker hit credit, with temporal (C05-eT) or constant (C05-eC) encoder credit fixed (n = 5 pairs per row). The eT/eC suffix identifies the fixed encoder setting; it has a different role from wT/wC in D and E. Online visited-cell fraction and amplitude-weighted DG spatial score are averaged over 90–99 million training frames. Mean temporal-minus-constant effects are +0.026/+0.013 for C05-eT and −0.087/+0.023 for C05-eC (coverage/DG score). The right two plots come from a separate episode-long, orthogonal-FiLM goal study, not the C15 checkpoints in A–E. They compare longer PPO credit, which jointly raises γ from 0.99 to 0.999 and GAE λ from 0.95 to 0.99, against the original settings at 100 million frames. Coverage AUC rises by +14.2 points for P and +11.9 for C15 on average (n = 3 matched seeds each); mean learned-unit DG spatial information on separate 10,000-decision policy-driven frozen rollouts falls by 0.022 and 0.056 bits, respectively. The long-credit frozen paths differ between conditions, so this is not a fixed-input representation contrast. The four F axes are restricted to the observed absolute-value ranges rather than starting at zero. Coverage and detector hits are not executed goal control.